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SEMICONDUCTOR DEVICE INCLUDING A TEST LINE PORTION AND METHOD OF MAKING AND TESTING THE SAME

Abstract

A semiconductor device includes a first die including a first die edge and a first die seal ring, a second die bonded to the first die and including a second die edge and a second die seal ring, and a test line portion including at least one of a first die test line portion in the first die between the first die edge and the first die seal ring, or a second die test line portion in the second die between the second die edge and the second die seal ring.

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Background/Summary

BACKGROUND

[0001] A die (e.g., semiconductor die) may include a seal ring also known as a scribe line seal or kerf seal. The seal ring may serve multiple purposes and functions. The seal ring may serve as a boundary around the die, enclosing active circuitry and other components as well as protecting the circuitry from external contaminants, moisture, and mechanical damage. The seal ring may also provide electrical and thermal isolation between the active circuitry on the die and components outside the seal ring. The seal ring may also add structural strength to the die and thereby inhibit warping or cracking of the die.

[0002] The seal ring may also include alignment marks or fiducials that aid in the precise positioning of the die during assembly and packaging and thereby improve the overall quality and reliability of the device including the die. The seal ring may also provide scribe lines for guiding a dicing tool to make precise cuts in separating the die from a wafer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale and are used for illustration purposes only. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. **1**A is a vertical cross-sectional view of a semiconductor device according to one or more embodiments.

[0005] FIG. **1**B is a schematic illustration of the semiconductor device in a top-down (z-direction) view according to one or more embodiments.

[0006] FIG. **1**C is an exploded perspective view of the semiconductor device according to one or more embodiments.

[0007] FIG. **2**A is an intermediate structure including the first bonding film on the carrier substrate according to one or more embodiments.

[0008] FIG. **2**B is an intermediate structure including the first die and third die on the first bonding film according to one or more embodiments.

[0009] FIG. **2**C is an intermediate structure including the first encapsulation layer on the first bonding film according to one or more embodiments.

[0010] FIG. **2**D is an intermediate structure including the second layer on a second carrier substrate according to one or more embodiments.

[0011] FIG. **2**E is an intermediate structure including the second layer on the first layer according to one or more embodiments.

[0012] FIG. **2**F is an intermediate structure after removal of the second carrier substrate according to one or more embodiments.

[0013] FIG. **2**G is an intermediate structure including a multilayer opening in the passivation layer and the second bonding film according to one or more embodiments.

[0014] FIG. **2**H is an embodiment structure including the metal bump on the die bonding pad in the multilayer opening according to one or more embodiments.

[0015] FIG. **3** is a flow chart illustrating a method of forming a semiconductor device according to one or more embodiments.

[0016] FIG. **4** is a schematic illustration of a method of testing the semiconductor device according to one or more embodiments.

[0017] FIG. **5**A is a vertical cross-sectional view of the semiconductor device having a first alternative design according to one or more embodiments.

[0018] FIG. **5**B is a schematic illustration of a method of testing the semiconductor device having the first alternative design according to one or more embodiments.
[0019] FIG. **6** is a flow chart illustrating a method of testing the semiconductor device according to one or more embodiments.
[0020] FIG. **7** is a vertical cross-sectional view of the semiconductor device having a second alternative design according to one or more embodiments.
[0021] FIG. **8** is a vertical cross-sectional view of the semiconductor device having a third alternative design according to one or more embodiments.

DETAILED DESCRIPTION

[0022] The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific embodiments or examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, dimensions of elements are not limited to the disclosed range or values, but may depend upon process conditions and/or desired properties of the device. Moreover, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed interposing the first and second features, such that the first and second features may not be in direct contact. Various features may be arbitrarily drawn in different scales for simplicity and clarity.
[0023] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The tool may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.
[0024] A related die (e.g., semiconductor die, a stack of integrated semiconductor chips, chiplet, etc.) such as a central processing unit (CPU) die or a static random access memory (SRAM) die may include a seal ring design. In such related dies, there may be no metal pattern or devices located outside the seal ring (i.e., between the die edge and the seal ring). In these related dies, an amount of the die outside the seal ring may be substantially uniform around the outer periphery of the die. That is, a variation in the distance between the die edge and the seal ring may be within about 10% (i.e., no greater than about 10%).
[0025] At least one embodiment of the present disclosure may include a novel semiconductor device including a test line portion (e.g., die overhang test line portion). At least one embodiment may include the test line portion in at least one die (e.g., semiconductor die) of a die module including a plurality of stacked dies (e.g., chiplet stack) to provide design flexibility. The test line portion may be located between a seal ring in the die and an edge of the die (e.g., the die edge).
[0026] The test line portion may include a test line electrical connection design that may allow for tool testing. In at least one embodiment, the test line portion may be configured to permit testing of the semiconductor device such as by a wafer acceptance testing (WAT) device. In at least one embodiment, the test line portion may be used to monitor an influence of process (e.g., manufacturing process) on the semiconductor device (e.g., an influence on an integrated circuit in the semiconductor device). In at least one embodiment, the test line portion may be used to monitor a charging damage effect on the semiconductor device. The test line portion may be observable from a cross section of the semiconductor device.
[0027] In at least one embodiment, the test line portion may be included in a die overhang (e.g., a portion of the die outside the die seal ring) and referred to as a die overhang test line portion. The die overhang test line portion may include the same substrate (e.g., semiconductor substrate; silicon

substate) as the rest of the die. In the die overhang test line portion, a distance between the seal ring and the die edge may be greater than in other parts of the die. The die overhang test line portion may include a metal layer pattern. In at least one embodiment, the die overhang test line portion may include test line outside the die seal ring surrounding a functional circuit portion of the die. The die overhang test line portion may serve as a process monitor. The die overhang test line portion may or may not include its own seal ring (i.e., a seal ring outside of the die seal ring).

[0028] The test line portion may provide an innovative design. A test site for the test line portion may be provided by performing a photolithographic process. In the photolithographic process, a passivation layer may be etched by an etching process (e.g., a Pass2_etch) through openings in a patterned photoresist mask to expose the test site. In at least one embodiment, the etching process may expose a metal pad (e.g., open an aluminum pad) that may be used for tool testing such as WAT tool testing.

[0029] In at least one embodiment, the semiconductor device may include a first die and a second die bonded to the first die. The semiconductor device may also include a substrate and the first die may be located in a first level on the substrate. The second die may be located in a second level adjacent the first level and have a width greater than a width of the first die. A third die may also be located adjacent the first die in the first level. The third die may include a dummy die (e.g., non-functioning die).

[0030] In at least one embodiment, the second die may include the second die test line portion. The second die may also be electrically coupled to the first die by an electrical connection (e.g., by a through silicon via (TSV)). The second die test line portion in the second die may, therefore, be used as a test line. In particular, the second die test line portion in the second die may, therefore, be used to monitor the influence of the process (e.g., manufacturing process) on the semiconductor device (e.g., the first die and the second die) and the integrated circuit in the semiconductor device. In particular, a test site may be formed in the second die by an etching process (e.g., a Pass2_etch). The test site may include, for example, a metal pad (e.g., aluminum pad) expose during the etching process. In particular, the metal pad may be contacted by one or more probes of a testing tool (e.g., a WAT testing tool).

[0031] In at least one embodiment, the first die may include the first die test line portion. The first die may also be electrically coupled to the second die by an electrical connection (e.g., by a TSV). The first die test line portion in the first die may, therefore, be used as a test line. In particular, the first die test line portion in the first die may be used to monitor the influence of the process on the semiconductor device (e.g., the first die and the second die) and the integrated circuit in the semiconductor device. The test site may again be formed in the second die as described above and electrically coupled to the first die test line portion in the first die.

[0032] In at least one embodiment, the first die may include a first die test line portion. However, the first die test line portion may not be electrically coupled to the first die and/or the second die. In this case, the first die test line portion may not be used for testing, but may be used, for example, as a heat dissipation structure. The first die test line portion may also be used to reduce the area of the third die. In at least one embodiment, the first die test line portion may completely eliminate the need for the third die. This may help to reduce cost and balance stress (e.g., stress caused by mismatched coefficient of thermal expansions (CTEs)) in the level of the first die (e.g., same level, same structure).

[0033] In at least one embodiment, the first die may include a first die test line portion and the second die may include a second die test line portion. This may allow for separate monitoring of the first die and the second die. In particular, this may allow the first die and the second die to be easily analyzed by separate WAT testing. Separate monitoring may be especially advantageous where the first die and the second die are formed in different processes. Electrical die sorting (EDS) may affect the affordability of the die module.

[0034] Referring to the drawings, FIG. **1**A is a vertical cross-sectional view of a semiconductor

device **100** according to one or more embodiments. The semiconductor device **100** may include a die stack (e.g., a chiplet stack) including a plurality of dies (e.g., semiconductor dies). As illustrated in FIG. **1**A, the semiconductor device **100** may include a first layer **10** including a first encapsulation layer **118** and a first die **110** in the first encapsulation layer **118**. The semiconductor device **100** may also include a second layer **20** including a second encapsulation layer **128** and a second die **120** in the second encapsulation layer **128**. The second die **120** may be electrically coupled to the first die **110**. The first die **110** and second die **120** may together form an integrated circuit in the semiconductor device **100**.
[0035] As further illustrated in FIG. **1**A, the first die **110** and second die **120** may be arranged in the semiconductor device **100** with a front-to-back arrangement. In particular, a front side **120***f* of the second die **120** may be bonded to a backside **110***b* of the first die **110**. A backside **120***b* of the second die **120** may face away from the first die **110** and a frontside **110***f* of the first die **110** may face away from the second die **120**.
[0036] In at least one embodiment, the first die **110** and/or the second die **120** may include a second die test line portion **180-2**. The second die test line portion **180-2** may provide for the testing of the semiconductor device **100** (e.g., allow monitoring an effect of processing on the semiconductor device **100**).
[0037] As further illustrated in FIG. **1**A, the first die **110** may include a die substrate **108**. The die substrate **108** may include a semiconductor material such as silicon, germanium, silicon germanium, etc. In at least one embodiment, the first die **110** may be built upon a silicon wafer in wafer level processing. The first die **110** may be formed by dicing the silicon wafer into a plurality of dies including the first die **110**. The first die **110** may, therefore, include a portion of silicon wafer as the die substrate **108**. The dicing of the silicon wafer may also form a first die edge **110***a* of the first die **110**. The first die edge **110***a* may be formed around an entire periphery of the first die **110**. One or more active devices may be formed in and/or on the die substrate **108**. The active devices may include one or more gate electrodes **109** formed on the die substrate **108**. The gate electrodes **109** may be formed of a conductive material such as metal, polysilicon, etc.
[0038] The first die **110** may also include one or more interlayer dielectric (ILD) layers **112** on the die substrate **108**. The gate electrodes **109** may be located on the die substrate **108** in the ILD layers **112**. The ILD layers **112** may include, for example, undoped silicon glass (USG), fluorosilicate glass (FSG), silicon oxide (Si.sub.xO.sub.y), hafnium silicate (HfSiO.sub.4), zirconium silicate (ZrSiO.sub.4), tetraethyl orthosilicate (TEOS), hydrogen silsesquoxane (HSQ), etc. Other suitable dielectric materials may be used for the ILD layers **112**. The ILD layers **112** may be formed by a suitable deposition method such as chemical vapor deposition (CVD), plasma-enhanced CVD (PECVD), low pressure chemical vapor deposition (LPCVD), physical vapor deposition (PVD), atomic layer deposition (ALD), spin-on coating, or lamination.
[0039] One or more intermetal dielectric (IMD) layers **114** may be formed on the ILD layers **112**. The IMD layers **114** may also include, for example, undoped silicon glass (USG), fluorosilicate glass (FSG), silicon oxide, hafnium silicate (HfSiO.sub.4), zirconium silicate (ZrSiO.sub.4), tetraethyl orthosilicate (TEOS), hydrogen silsesquoxane (HSQ), etc. Other suitable dielectric materials may be used for the IMD layers **114**. The IMD layers **114** may be separated by one or more etch stop layers **114***a*. The etch stop layers **114***a* may include, for example, silicon nitride (SixNy), silicon carbide, etc. Other suitable materials may be used for the etch stop layers **114***a*. The IMD layers **114** and etch stop layers **114***a* may be formed by a suitable deposition method such as CVD, PVD, PECVD, LPCVD, ALD, spin-on coating, lamination, etc.
[0040] The first die **110** may also include one or more metal features **116** in the IMD layers **114**. The metal features **116** may be electrically connected to the gate electrodes **109**. The metal features **116** may include, for example, metal vias and metal traces (e.g., metal lines). The metal vias may be formed between the metal traces and interconnect the metal traces in the various IMD layers **114**. The metal features **116** may be formed of suitable metals such as copper, copper alloys,

aluminum, aluminum alloys, or some combination thereof. Other suitable conductive metal materials for use as the metal features **116** are within the contemplated scope of disclosure.

[0041] Referring to FIGS. **1**A and **1**B, the first die **110** may also include a first die seal ring **117-1** extending through the IMD layers **114** and into the ILD layers **112**. The first die seal ring **117-1** may contact the substrate **108**. The first die seal ring **117-1** may be electrically isolated from the metal features **116** and formed so as to surround (e.g., encircle) a functional circuit portion **170** of the first die **110**. The first die seal ring **117-1** may provide protection for the features of the first die **110** from water, chemicals, residue, and/or contaminants that may be present during the processing of the first die **110**. The first die seal ring **117-1** may be formed of a conductive material (e.g., metal material) and more particularly, may be formed of the same material, at the same time, and by the same process as the metal features **116**. More particularly, the first die seal ring **117-1** may include conductive lines and via structures that are connected to each other, and may be formed simultaneously with the metal lines and conductive vias of the metal features **116**. For example, the first die seal ring **117-1** may include copper at an atomic percentage greater than 80%, such as greater than 90% and/or greater than 95% although greater or lesser percentages may be used.

[0042] In at least one embodiment, the metal features **116** and/or the first die seal ring **117-1** may be formed by a dual-Damascene process or by multiple single Damascene processes. A single-Damascene process may generally form and fill a single feature with copper per Damascene stage. A dual-Damascene process may generally form and fill two features with a metal (e.g., copper) at once in a single step. For example, a trench and overlapping through-hole may both be filled with a single metal deposition using a dual-Damascene process. In alternative embodiments, the metal features **116** and/or the first die seal ring **117-1** may be formed by an electroplating process.

[0043] In at least one embodiment, the metal features **116** and/or the first die seal ring **117-1** may be formed by a Damascene process in which the IMD layers **114** are patterned (e.g., by a photolithographic process) to form openings such as trenches and/or though-holes (e.g., via holes). A metal layer (e.g., copper layer) may then be deposited by a suitable deposition method such as CVD, PVD, PECVD, LPCVD, ALD, spin-on coating, lamination, etc. A planarization process, such as chemical-mechanical planarization (CMP) may then be performed to remove excess metal (e.g., overburden).

[0044] The patterning, metal deposition, and planarizing processes may be performed separately for each dielectric layer of the IMD layers **114** in order to form an interconnect structure made up of the metal features **116**. For example, a dielectric layer of the IMD layers **114** may be deposited on the ILD layers **112** by a suitable deposition method such as CVD, PVD, PECVD, LPCVD, ALD, spin-on coating, lamination, etc. One or more openings may then be formed in the dielectric layer. The openings may be formed, for example, using a photolithographic process. In that process, a photoresist mask (not shown) may be formed on an upper surface of the ILD layers **112**. The photoresist mask may be photolithographically patterned to include one or more openings. An etching process (e.g., wet etching, dry etching, etc.) may then be used to form one or more openings in the dielectric layer through the openings in the photoresist mask. The photoresist mask may be subsequently removed by ashing, dissolving the photoresist mask or by consuming the photoresist mask during the etch process.

[0045] A suitable deposition method (e.g., CVD, PVD, PECVD, LPCVD, ALD, spin-on coating, lamination, etc.) may then be performed to fill the openings in the dielectric layer of the IMD layers **114**. A planarization process may then be performed to remove the overburden and form the metal features **116** in the dielectric layer. These process steps may be repeated to form the IMD layers **114** and the corresponding metal features **116** and first die seal ring **117-1** and thereby complete the interconnect structure and first die seal ring **117-1**.

[0046] A passivation layer **119** may be formed on the IMD layers **114**. In at least one embodiment, the passivation layer **119** may include silicon oxide, silicon nitride, benzocyclobutene (BCB) polymer, polyimide (PI), polybenzoxazole (PBO) or a combination thereof. Other suitable dielectric materials are within the contemplated scope of disclosure. The passivation layer **119** may be formed by a suitable suitable deposition method such as CVD, PVD, PECVD, LPCVD, ALD, spin-on coating, lamination, etc.

[0047] One or more contact pads **103** may be formed through an opening in the passivation layer **119** onto the metal features **116** and the first die seal ring **117-1**. The contact pads **103** may include a metal such as aluminum, copper, etc. The contact pads **103** may be formed by performing a photolithographic process (similar to the photolithograpic process described above for forming the metal features **116** in the first die **110**) to form the opening, and depositing a metal layer in the opening by a suitable deposition process such as CVD, PVD, PECVD, LPCVD, ALD, spin-on coating, lamination, etc.

[0048] A gap fill dielectric layer **104** may then be formed on the passivation layer **119** and over and around the contact pads **103**. The gap fill dielectric layer **104** may be formed of the same material as the IMD layers **114** (e.g., undoped silicon glass (USG), fluorosilicate glass (FSG), silicon oxide, hafnium silicate (HfSiO.sub.4), zirconium silicate (ZrSiO.sub.4), tetraethyl orthosilicate (TEOS), hydrogen silsesquoxane (HSQ), etc. The gap fill dielectric layer **104** may be formed by a suitable deposition process such as CVD, PVD, PECVD, LPCVD, ALD, spin-on coating, lamination, etc.

[0049] A backside etch stop layer **113** may be formed on the gap fill dielectric layer **104**. The backside etch stop layer **113** may include, for example, silicon nitride (SixNy), silicon carbide, etc. Other suitable materials may be used for the backside etch stop layer **113**. The backside etch stop layer **113** may be formed by a suitable deposition method such as CVD, PVD, PECVD, LPCVD, ALD, spin-on coating, lamination, etc.

[0050] A die bonding film **105** (e.g., hybrid bonding film) may be formed on the backside etch stop layer **113**. The die bonding film **105** may include an oxide such as silicon oxide. The die bonding film **105** may be formed by a suitable deposition process such as CVD, PVD, etc. One or more die bonding pads **107** may then be formed through a multilayer opening including an opening in the die bonding film **105**, an opening in the backside etch stop layer **113**, an opening in the gap fill dielectric layer **104** and an opening in the passivation layer **119**, and contact a metal feature **116** in the IMD layers **114**. The die bonding pads **107** may be formed of a material similar to the material of the metal features **116**. The die bonding pads **107** may be formed by performing a photolithographic process (similar to the photolithograpic process described above for forming the metal features **116** in the first die **110**) to form the multilayer opening including the openings in the die bonding film **105**, backside etch stop layer **113**, gap fill dielectric layer **104** and passivation layer **119**, and depositing a metal layer in the multilayer opening by a suitable deposition process such as CVD, PVD, PECVD, LPCVD, ALD, spin-on coating, lamination, etc.

[0051] The first layer **10** of the semiconductor device **100** may also include a third die **130** in the first encapsulation layer **118**. The third die **130** may include, for example, a dummy die (e.g., non-functional die). The third die **130** may have a thickness in the z-direction that is substantially the same as a thickness of the first die **110**. The third die **130** may include, for example, a silicon die. Other suitable dielectric materials are within the contemplated scope of disclosure.

[0052] The first encapsulation layer **118** may be formed on the first die **110** and third die **130** so as to encapsulate at least a portion of the first die **110** and at least a portion of the third die **130**. The first encapsulation layer **118** may include an organic material such as molding compound material (e.g., formed through molding), or an inorganic material (e.g., formed through PVD). The first encapsulation layer **118** may include, for example, silicon dioxide. The first encapsulation layer **118** may alternatively include undoped silicon glass (USG), fluorosilicate glass (FSG), SiC, SiON, SiN, SiCN, a low-K film, an extreme low-K (ELK) film, phosphor-silicate glass (PSG) and tetra-ethoxy-silane (TEOS). Other dielectric materials for use as the first encapsulation layer **118** are within the contemplated scope of disclosure. The first encapsulation layer **118** may alternatively or additionally include one or more molding compounds such as an epoxy molding compound (EMC).

[0053] As further illustrated in FIG. 1A, the first layer of the semiconductor device **100** may include a first bonding film **141** (e.g., fusion bonding film). The semiconductor device **100** may also include a first carrier substrate **102**. The first carrier substrate **102** may be bonded to the first layer **10** (e.g., the first die **110**, third die **130** and first encapsulation layer **118**) by the first bonding film **141**. A function of the first bonding film **141** may include bonding to the first die **110** and third die **130** and bonding the first die **110** and third die **130** to the first carrier substrate **102**. The first bonding film **141** may include, for example, silicon oxynitride, silicon oxide, etc. Other suitable dielectric materials for the first bonding film **141** are within the contemplated scope of disclosure.
[0054] The first layer **10** of the semiconductor device **100** may also include a first die bonding layer **143**. The first die **110** may be bonded to the first bonding film **141** by the first die bonding layer **143**. The first die bonding layer **143** may have a width substantially the same as a width of the first die **110**. The first die bonding layer **143** may have a thickness substantially the same as a thickness of the first bonding film **141**.
[0055] The first layer **10** of the semiconductor device **100** may also include a third die bonding layer **144**. The third die **130** may be bonded to the first bonding film **141** by the third die bonding layer **144**. The third die bonding layer **144** may have a width substantially the same as a width of the third die **130**. The third die bonding layer **144** may have a thickness substantially the same as a thickness of the first bonding film **141**.
[0056] Each of the first die bonding layer **143** and the third die bonding layer **144** may include, for example, silicon oxynitride, silicon oxide, etc. Other suitable dielectric materials for the first die bonding layer **143** and the third die bonding layer **144** are within the contemplated scope of disclosure.
[0057] An alignment mark **145** may be located in the first bonding film **141**. The alignment mark **145** may be located over a portion of the first encapsulation layer **118**. The alignment mark **145** may have a width less than a width of the portion of the first encapsulation layer **118**. The alignment mark **145** may have a thickness substantially the same as a thickness of the first bonding film **141**. In at least one embodiment, the alignment mark **145** may include a die-to-wafer alignment mark. The alignment mark **145** may include a metal such as copper. Other suitable materials may be used in the alignment mark **145**.
[0058] The second die **120** may have a structure that is substantially similar to the structure of the first die **110**. The second die **120** may also be formed by a method that is substantially similar to the method of forming the first die **110**. Similar to the first die **110**, the second die **120** may be built upon a silicon wafer in wafer level processing. The second die **120** may be formed by dicing the silicon wafer into a plurality of dies including the second die **120**. The second die **120** may, therefore, include a portion of silicon wafer as the die substrate **108**. The dicing of the silicon wafer may also form a second die edge **120***a* of the second die **120**. The second die edge **120***a* may be formed around an entire periphery of the second die **120**.
[0059] The second die **120** may further include one or more active devices in and/or on the die substrate **108** including one or more gate electrodes **109**. The second die **120** may also include one or more interlayer dielectric (ILD) layers **112** on the die substrate **108** and one or more intermetal dielectric (IMD) layers **114** on the ILD layers **112**. The IMD layers **114** may be separated by one or more etch stop layers **114***a*.
[0060] The second die **120** may also include one or more metal features **116** and a second die seal ring **117-2** in the IMD layers **114**. The second die seal ring **117-2** may have a structure and function substantially similar to a structure and function of the first die seal ring **117-1** in the first die **110**. The second die seal ring **117-2** may be electrically isolated from the metal features **116** and formed so as to surround (e.g., encircle) a functional circuit portion **170** of the second die **120**.
[0061] The second die **120** may additionally include one or more through silicon vias (TSV) **129** (e.g., second die vias) contacting one or more of the metal features **116**. The TSVs **129** may extend through at least a portion of the IMD layers **114**, the ILD layers **112** and the die substrate **108**. A

surface of the TSVs **129** may be substantially coplanar with a surface of the die substrate **108** in the second die **120**. The TSVs **129** may include, for example, copper, gold, silver, aluminum or the like, or an alloy of these metals such as aluminum copper (AlCu) alloy. Other suitable materials for use in the TSVs **129** are within the contemplated scope of disclosure.

[0062] The second die **120** may also include the passivation layer **119** on the IMD layers **114**, and one or more contact pads **103** may be formed through an opening in the passivation layer **119** onto the metal features **116** and the second die seal ring **117-2**. The second die **120** may also include the gap fill dielectric layer **104** on the passivation layer **119** and over and around the contact pads **103**. The second die **120** may also include the backside etch stop layer **113** on the gap fill dielectric layer **104**.

[0063] The second encapsulation layer **128** may be formed on the second die **120** so as to encapsulate at least a portion of the second die **120**. The second encapsulation layer **128** may include an organic material such as molding compound material (e.g., formed through molding), or an inorganic material (e.g., formed through PVD). The second encapsulation layer **128** may include, for example, silicon dioxide. The second encapsulation layer **128** may alternatively include undoped silicon glass (USG), fluorosilicate glass (FSG), SiC, SiON, SiN, SiCN, a low-K film, an extreme low-K (ELK) film, phosphor-silicate glass (PSG) and tetra-ethoxy-silane (TEOS). Other dielectric materials for use as the second encapsulation layer **128** are within the contemplated scope of disclosure. The second encapsulation layer **128** may alternatively or additionally include one or more molding compounds such as an epoxy molding compound (EMC).

[0064] The second layer **20** of the semiconductor device **100** may also include a die bonding film **105** and one or more die bonding pads **107** in the die bonding film **105**. The die bonding pads **107** may contact one of the metal contact pads **103** on a metal feature **116**. The second die **120** and second encapsulation layer **128** may be located on the die bonding film **105**. The second layer **20** may further include a passivation layer **138**. The die bonding film **105** may be located on the passivation layer **138**. The passivation layer **138** may include, for example, silicon nitride, undoped silicate glass (USG) or silicon dioxide. Other materials for the passivation layer **138** are within the contemplated scope of disclosure. The second layer **20** may further include a second bonding film **142** (e.g., fusion bonding film). The passivation layer **138** may be located on the second bonding film **142**. A function of the second bonding film **142** may include bonding elements of the second layer **20** (e.g., the second die **120**) to the second carrier substrate **202**. The second bonding film **142** may be formed of the same materials as the first bonding film **141** and may be formed using the same process as the first bonding film **141**. The second bonding film **142** may include, for example, silicon oxynitride or silicon dioxide. Other suitable dielectric materials for the second bonding film **142** are within the contemplated scope of disclosure. One or more metal bumps **190** may be formed in the passivation layer **138** and second bonding film **142** so as to contact the die bonding pads **107** (in the die bonding film **105**) that are connected to the metal features **116** of the second die **120**. The metal bumps **190** may be used to electrically couple the semiconductor device **100** to a separate substrate.

[0065] The second layer **20** of the semiconductor device **100** may also include a bonding layer **150**. The bonding layer **150** may include a hybrid bonding film and serve as a hybrid bond interface. The second layer **20** of the semiconductor device **100** may be bonded to the first layer **10** of the semiconductor device **100** by the bonding layer **150**. The bonding layer **150** may extend the entire width of the semiconductor device **100** in the x-direction. The bonding layer **150** may also include, for example, silicon oxynitride, silicon oxide, etc. Other suitable dielectric materials for the bonding layer **150** are within the contemplated scope of disclosure. The bonding layer **150** may include a bonding layer bonding pad **157** connected to the TSV **129** in the second die **120** and to the die bonding pad **107** in the first die **110**. The bonding layer bonding pad **157** may be formed of the same materials as the die bonding pad **107**. Other suitable materials are within the contemplated scope of disclosure. In at least one embodiment, a bond between the first layer **10** and the second layer **20** may include a hybrid bond (e.g., oxide bond, metal bond) including the bonding layer **150** and the bonding layer bonding pad **157**.

[0066] A redistribution layer structure (not shown) may optionally be formed in the bonding layer **150**. The optional redistribution layer structure may be used to interconnect the dies in the first layer **10** (e.g., connect the first die **110** to other dies in the first layer **10**). The optional redistribution layer structure may also be used to connect the dies in the first layer **10** (e.g., first die **110**) to the dies in the second layer **20** (e.g., second die **120**).

[0067] As illustrated in FIG. **1**A, the semiconductor device **100** may include a sidewall **100***a* that extends along an entire height of the semiconductor device **100** from the second bonding film **142** to the first carrier substrate **102**. The sidewall **100***a* may include primarily a sidewall **128***a* of the second encapsulation layer **128**, a sidewall **118***a* of the first encapsulation layer **118***a* and a sidewall **102***a* of the first carrier substrate **102**. The sidewall **100***a* may also include an end of the second bonding film **142**, an end of the passivation layer **138**, an end of the bonding layer **150**, and an end of the first bonding film **141**.

[0068] The second die test line portion **180-2** may be integrally formed as one continuous unit with the remainder of the second die **120**. The second die test line portion **180-2** may be located adjacent the functional circuit portion **170** of the second die **120**. The second die test line portion **180-2** may be bounded on one lateral side (e.g., in the x-y plane) by the functional circuit portion **170** and bounded on the three remaining lateral sides by the second die edge **120***a*. In at least one embodiment, the second die test line portion **180-2** may be located in a region of the second die **120** where a distance between the second die seal ring **117-2** and the die edge **120***a* may be greater than in other parts of the second die **120**.

[0069] The dashed line in FIG. **1**A may be considered an imaginary boundary between the functional circuit portion **170** and the second die test line portion **180-2**. The second die **120** may not be separated at the boundary but may be continuously formed across the boundary. The substrate **108** included in the functional circuit portion **170** may extend into and form part of the second die test line portion **180-2**, the ILD layers **112** included in the functional circuit portion **170** may extend into and form part of the second die test line portion **180-2**, and so on.

[0070] The second die test line portion **180-2** may include one or more test line portion metal features **116**T that are substantially similar to the metal features **116** in the functional circuit portion **170**. The test line portion metal features **116**T may have substantially the same structure and composition as the metal features **116** in the functional circuit portion **170**. The test line portion metal features **116**T may be formed concurrently with the metal features **116** and in the same processing steps as the metal features **116**.

[0071] The second die test line portion **180-2** in the second die **120** may have a configuration associated with the configuration of the functional circuit portion **170** of the second die **120**. In at least one embodiment, the second die test line portion **180-2** may include a metal pattern that is substantially the same as a metal pattern in the functional circuit portion **170** of the second die **120**.

[0072] The test line portion metal features **116**T may be electrically coupled to the metal features **116** of the function circuit portion **170**. The second die test line portion **180-2** may be electrically coupled to the first die **110** through the functional circuit portion **170** of the second die **120**. In particular, the second die test line portion **180-2** may be electrically coupled to the first die **110** through the metal features **116** and the TSV **129** of the functional circuit portion **170** and the bonding layer bonding pad **157** in the bonding layer **150**.

[0073] The second die test line portion **180-2** may include one or more test line portion seal rings **117**T that are substantially the same structure and composition as the second die seal ring **117-2** in the functional circuit portion **170**. The test line portion seal rings **117**T may have a substantially similar structure as the second die seal ring **117-2**. The test line portion seal rings **117**T may be formed concurrently with the second die seal ring **117-2** and in the same processing steps as the second die seal ring **117-2**.

[0074] The second die seal ring **117-2** and/or the test line portion seal rings **117**T may also include alignment marks or fiducials that aid in the precise positioning of the second die **120** during assembly and packaging and thereby improve the overall quality and reliability of the semiconductor device **100**. The second die seal ring **117-2** and/or the test line portion seal rings **117**T may also provide scribe lines for guiding a dicing tool to make precise cuts in separating the second die **120** from other semiconductor devices formed on the first carrier substrate **102**.
[0075] The second die test line portion **180-2** may include a test line electrical connection design that may allow for tool testing of the semiconductor device **100**. In at least one embodiment, the second die test line portion **180-2** may be configured to permit testing of the semiconductor device **100** (e.g., testing of all of the functional dies in the semiconductor device **100**) such as by a wafer acceptance testing (WAT) device. In at least one embodiment, second die test line portion **180-2** may be used to monitor an influence of processing on the semiconductor device **100**. In at least one embodiment, the second die test line portion **180-2** may be used to monitor a charging damage effect on the semiconductor device **100**.
[0076] FIG. **1**B is a schematic illustration of the semiconductor device **100** in a top-down (z-direction) view according to one or more embodiments. The schematic illustration in FIG. **1**B illustrates a relative lateral location (e.g., in the x-y plane) of the first die **110**, second die **120**, third die **130** and first carrier substrate **102** in the semiconductor device **100**. The schematic illustration in FIG. **1**B further illustrates a relative lateral location of the first die seal ring **117-1**, the second die seal ring **117-2** and the test line portion seal ring **117**T in the semiconductor device **100**. The first encapsulation layer **118** and second encapsulation layer **128** are omitted from FIG. **1**B for ease of understanding. The vertical cross-sectional view in FIG. **1**A is across the line A-A′ in the schematic illustration of FIG. **1**B.
[0077] As illustrated in FIG. **1**B, an area of the first die **110** may be less than an area of the third die **130**. A length in the x-direction of the die **110** may be less than a length in the x-direction of the first carrier substrate **102**. A width in the y-direction of the second die **120** may be less than a width in the y-direction of the first carrier substrate **102**.
[0078] An area of the second die **120** may be greater than a combined area of the first die **110** and the third die **130**. A combined length in the x-direction of the first die **110** and the third die **130** may be less than a length in the x-direction of the second die **120**. A greatest width in the y-direction of the first die **110** and third die **130** may be less than a width in the y-direction of the second die **120**. In at least one embodiment, each of the first die **110** and third die **130** may be located laterally within the area of the second die **120** (e.g., within the second die edge **120***a*).
[0079] An area of the first carrier substrate **102** may be greater than the area of the second die **120**. A length in the x-direction of the second die **120** may be less than a length in the x-direction of the first carrier substrate **102**. A width in the y-direction of the second die **120** may be less than a width in the y-direction of the first carrier substrate **102**. In at least one embodiment, the second die **120** may be located laterally within the area of the first carrier substrate **102**.
[0080] As further illustrated in FIG. **1**B, the first die **110** may be located within the second die seal ring **117-2**. The third die **130** may include a first portion located over the functional circuit portion **170** of the second die **120** and a second portion located over the second die test line portion **180-2** of the second die **120**. The second die test line portion **180-2** may include a plurality of test line portion seal rings **117**T and a plurality of test line portion metal features **116**T. The plurality of test line portion metal features **116**T may be located within the test line portion seal rings **117**T and/or outside the test line portion seal rings **117**T.
[0081] FIG. **1**C is an exploded perspective view of the semiconductor device **100** according to one or more embodiments. The first bonding film **141**, bonding layer **150**, passivation layer **138**, second bonding film **142** and metal bump **190** are omitted from FIG. **1**C for ease of explanation.
[0082] The dashed line in the z-direction in FIG. **1**C is used to identify a centerpoint of the semiconductor device **100**. The centerpoint of each of the first layer **10**, second layer **20** and first

carrier substrate **102** may be substantially aligned with the centerpoint of the semiconductor device **100**. In at least one embodiment the sidewall **118***a* of the first encapsulation layer **118** may substantially aligned with the sidewall **128***a* of the second encapsulation layer **128** and with the sidewall **102***a* of the first carrier substrate **102** around an entire periphery of the semiconductor device **100**.

[0083] In the first layer **10** of the semiconductor device **100**, the first encapsulation layer **118** may surround an entire periphery of the first die **110** and an entire periphery of the third die **130**. The first layer **10** may include a substantially uniform upper surface. An upper surface of the first die **110** may be constituted by an upper surface of the die substrate **108** in the first die **110**. In at least one embodiment, an upper surface of the first encapsulation layer **118** may be substantially coplanar with an upper surface of the die substrate **108** in the first die **110** and an upper surface of the third die **130**.

[0084] In the second layer **20** of the semiconductor device **100**, the second encapsulation layer **128** may surround an entire periphery of the second die **120**. The second layer **20** may include a substantially uniform upper surface. An upper surface of the second die **120** may be constituted by an upper surface of the die substrate **108** in the second die **120**. In at least one embodiment, an upper surface of the second encapsulation layer **128** may be substantially coplanar with an upper surface of the die substrate **108** in the second die **120**.

[0085] Generally, the semiconductor device **100** may be formed by forming the first layer **10** on the first carrier substrate **102**, and forming the second layer **20** on the first layer **10**. In at least one embodiment, the semiconductor device **100** may be formed in a process in which a plurality of semiconductor devices are formed concurrently on a carrier substrate. In that process, the first layer **10** may be formed by mounting a plurality of the first dies **110** and a plurality of the third dies **130** on the carrier substrate and encapsulating the first dies **110** and third dies **130** in the first encapsulation layer **118**. The second layer **20** may then be formed by mounting a plurality of the second dies **120** on the first layer **10** and encapsulating the second dies **120** in the second encapsulation layer **128**. A dicing step may then be performed to separate the plurality of semiconductor devices.

[0086] FIGS. **2**A-**2**H illustrate various intermediate structures in a method of forming the semiconductor device **100** according to one or more embodiments. FIG. **2**A is an intermediate structure including the first bonding film **141** on the first carrier substrate **102** according to one or more embodiments.

[0087] In at least one embodiment, the first carrier substrate **102** may be placed on a flat surface and the first bonding film **141** may be formed on an upper surface of the first carrier substrate **102**. The first bonding film **141** may be formed on the upper surface of the first carrier substrate **102**. In at least one embodiment, the first bonding film **141** may be formed by a depositing a bonding material by a suitable deposition method such as CVD, PVD, PECVD, LPCVD, ALD, spin-on coating, lamination, etc. The bonding material may then be planarized (e.g., by wet etching, drying etching, etc.) so as to form the first bonding film **141**.

[0088] The alignment mark **145** may then be formed in the first bonding film **141**. An opening for the alignment mark **145** may be formed in the first bonding film **141** by a photolithographic process. The photolithographic process may be similar to the photolithographic process described above for forming the metal features **116** in the first die **110**. In the photolithographic process, a photoresist mask (not shown) may be formed on a surface of the first bonding film **141**. The photoresist mask may be photolithographically patterned to include an opening. An etching process (e.g., wet etching, dry etching, etc.) may then be used to form an opening in the first bonding film **141** through the opening in the photoresist mask. The opening in the first bonding film **141** may be formed to expose a surface of the first carrier substrate **102**. The photoresist mask may be subsequently removed by ashing, dissolving the photoresist mask or by consuming the photoresist mask during the etch process.

[0089] A layer of metal material (e.g., copper) may then be deposited (e.g., by CVD, PVD or other suitable process) on the first bonding film **141** and in the opening in the first bonding film **141**. Excess metal material may then be removed (e.g., by CMP or other suitable planarizing process) so as to form the alignment mark **145** with a surface substantially coplanar with a surface of the first bonding film **141**.

[0090] FIG. **2**B is an intermediate structure including the first die **110** and third die **130** on the first bonding film **141** according to one or more embodiments. In at least one embodiment, the first die **110** may be placed on the first bonding film **141** by using an electromechanical pick-and-place (PNP) machine. The third die **130** may be placed on the first bonding film **141** separately from the first die **110** (e.g., in a separate step). The third die **130** may be placed on the first bonding film **141** adjacent to the first die **110** by using the PNP machine. The third die **130** may alternatively be placed on the first bonding film **141** first and the first die **110** placed on the first bonding film **141** adjacent the third die. The third die **130** may alternatively be placed on the first bonding film **141** concurrently with the first die **110** in a single step. In at least one embodiment, the first alignment mark **145** may be used to assist in properly aligning (e.g., locating, placing, etc.) the first die **110** and/or the third die **130**.

[0091] As illustrated in FIG. **2**B, the first die **110** and third die **130** may be placed on the first bonding film **141** so as to provide sufficient separation between the first die **110** and third die **130**. The separation may allow a sufficient quantity of the first encapsulation layer **118** to be formed in the gap between the first die **110** and third die **130** (see FIG. **1**C). The separation may also allow the first encapsulation film **118** to fill the gap (e.g., completely fill the gap) without leaving air pockets in the gap. The first die **110** and third die **130** may also be placed on the first bonding film **141** so that the combined length of the first die **110** and third die **130** in the x-direction is less than the length of the second die **120** in the x-direction (see FIG. **1**A).

[0092] FIG. **2**C is an intermediate structure including the first encapsulation layer **118** on the first bonding film **141** according to one or more embodiments. The first encapsulation layer **118** may be formed on the first bonding film **141** so as to encapsulate (e.g., at least partially encapsulate) the first die **110** and third die **130**. The first encapsulation layer **118** (e.g., an encapsulation material) may be deposited on the first bonding film **141**. The first encapsulation layer **118** may be deposited by a suitable deposition method such as CVD, PVD, PECVD, LPCVD, ALD, spin-on coating, lamination, etc.

[0093] In at least one embodiment, the first encapsulation layer **118** may be formed to have a height in the z-direction greater than a height of the first die **110** and greater than a height of the third die **130**. After the first encapsulation layer **118** has been formed, a planarization process (e.g., chemical mechanical polish (CMP)) may then be performed to planarize an upper surface of the first encapsulation layer **118**, an upper surface of the first die **110** and an upper surface of the third die **130**. The upper surface of the first encapsulation layer **118**, upper surface of the first die **110** and upper surface of the third die **130** may be made co-planar by the planarization process. The planarization process may also complete the formation of the first layer **10** (except for the dicing step to separate the semiconductor device **100**).

[0094] FIG. **2**D is an intermediate structure including the second layer **20** on a second carrier substrate **202** according to one or more embodiments. The second carrier substrate **202** may be substantially similar to the first carrier substrate **102**. The second layer **20** may be formed on the second carrier substrate **202** in a manner substantially similar to the manner in which the first layer **10** is formed on the first carrier substrate **102** (e.g., see FIGS. **2**A-**2**C).

[0095] An adhesive layer (not shown) may be applied to an upper surface of the second carrier substrate **202**. The second carrier substrate **202** may include an optically transparent material such as glass or sapphire. The adhesive layer may include, for example, a light-to-heat conversion (LTHC) layer or thermally decomposing adhesive material. Other suitable decomposable adhesive materials are within the contemplated scope of disclosure.

[0096] The second bonding film **142** may be formed on the adhesive layer on the upper surface of the second carrier wafer **202**. The second bonding film **142** may be formed on the upper surface of the second carrier substrate **202**. The second bonding film **142** may be formed in the same manner as the first bonding film **141**. In at least one embodiment, the second bonding film **142** may be formed by depositing a bonding material using a suitable deposition method such as CVD, PVD, PECVD, LPCVD, ALD, spin-on coating, lamination, etc. The bonding material may then be planarized (e.g., by wet etching, drying etching, etc.) so as to form the second bonding film **142**.
[0097] The passivation layer **138** may then be formed on the second bonding film **142**. The passivation layer **138** may be formed on the second layer **20**. The passivation layer **138** may be formed by depositing (e.g., by CVD, PVD or other suitable deposition technique) one or more layers of passivation material including silicon oxide, silicon nitride, low-k dielectric materials such as carbon-doped oxides, extremely low-k dielectric materials such as porous carbon doped silicon dioxide, a combination thereof or other suitable material. The passivation material may then be planarized (e.g., by wet etching, drying etching, etc.) so as to form the passivation layer **138**.
[0098] The die bonding film **105** may then be formed on the passivation layer **138**. In at least one embodiment, the die bonding film **105** may be formed by depositing a bonding material using a suitable deposition method such as CVD, PVD, PECVD, LPCVD, ALD, spin-on coating, lamination, etc. The bonding material may then be planarized (e.g., by wet etching, drying etching, etc.) so as to form the die bonding film **105**.
[0099] The second die **120** may then be placed on the die bonding film **105** by using an electromechanical PNP machine. The second die **120** may inverted so that a backside of the second die **120** contacts the die bonding film **105**. In particular, the backside etch stop layer **113** may be aligned with the die bonding film **105** so that an exposed contact pad **103** in the second die **120** contacts the die bonding pad **107** in the die bonding film **105**.
[0100] The second encapsulation layer **128** may then be formed on the second die **120** by a method similar to the method of forming the first encapsulation layer **118** described above. In particular, the second encapsulation layer **128** may be formed on the second die **120** and on the passivation layer **138** around the second die **120** so as to encapsulate (e.g., at least partially encapsulate) the second die **120**. The second encapsulation layer **128** (e.g., an encapsulation material) may be deposited on the second die **120** and the passivation layer **138**. The second encapsulation layer **128** may be deposited by a suitable deposition method such as CVD, PVD, PECVD, LPCVD, ALD, spin-on coating, lamination, etc.
[0101] In at least one embodiment, the second encapsulation layer **128** may be formed to have a height in the z-direction greater than a height of the second die **120**. After the second encapsulation layer **128** has been sufficiently formed, a planarization process (e.g., CMP) may then be performed to planarize a surface of the second encapsulation layer **128** and a surface of the second die **120**. The surface of the second encapsulation layer **128** and surface of the second die **120** may be made co-planar by the planarization process.
[0102] The bonding layer **150** may then be formed on the surface of the second encapsulation layer **128** and surface of the second die **120**. The bonding layer **150** may be formed in a manner substantially similar to the manner of forming the first bonding film **141** and the manner of forming the second bonding film **142**. In at least one embodiment, the bonding layer **150** may be formed on the surface of the second encapsulation layer **128** and surface of the second die **120**. In at least one embodiment, the bonding layer **150** may be formed by a suitable deposition method such as CVD, PVD, PECVD, LPCVD, ALD, spin-on coating, lamination, etc. In at least one embodiment, the bonding material may be deposited by CVD and a source of the bonding material may include a gas or liquid (e.g., tetraethyl orthosilicate (TEOS)).
[0103] The bonding layer bonding pad **157** may then be formed in the bonding layer **150**. An opening may be formed in the bonding layer **150** by a photolithographic process. The photolithographic process may be similar to the photolithographic process described above for

forming the metal features **116** in the first die **110**. In the photolithographic process, a photoresist mask (not shown) may be formed on a surface of the bonding layer **150**. The photoresist mask may be photolithographically patterned to include an opening over the TSV **129**. An etching process (e.g., wet etching, dry etching, etc.) may then be used to form an opening in the bonding layer **150** through the opening in the photoresist mask. The opening in the bonding layer **150** may be formed to expose a surface of the TSV **129**. The photoresist mask may be subsequently removed by ashing, dissolving the photoresist mask or by consuming the photoresist mask during the etch process.

[0104] A metal layer may then be formed on the bonding layer **150**, in the opening of the bonding layer **150** and onto the surface of the TSV **129**. The metal layer may be formed by a suitable deposition process such as CVD, PVD, PECVD, LPCVD, ALD, spin-on coating, lamination, etc. The metal layer may then be planarized (e.g., by CMP) so that a surface of the bonding layer bonding pad **157** is substantially co-planar with the surface of the bonding layer **150**. The forming of the bonding layer bonding pad **157** complete the formation of the second layer **20**.

[0105] FIG. **2**E is an intermediate structure including the second layer **20** on the first layer **10** according to one or more embodiments. In at least one embodiment, the second layer **20** may be bonded to the first layer **10** in a wafer-to-wafer bonding process.

[0106] The second layer **20** may be placed on first layer **10**, for example, by an electromechanical PNP machine. In particular, the intermediate structure in FIG. **2**D may be inverted and positioned over the first layer **10** so that the second layer **20** is between the first layer **10** and the second carrier substrate **202**. The second layer **20** may then be lowered onto the first layer **10** so that the bonding layer **150** in the second layer **20** contacts the first encapsulation layer **118**, the die bonding film **105** which is formed at the backside of the first die **110**, and the third die **130**. The second layer **20** may also be positioned on the first layer **10** so that the bonding layer bonding pad **157** in the bonding layer **150** contacts the die bonding pad **107** in the first die **110**.

[0107] The second layer **20** may also be positioned on the first layer **10** so that the second die **120** is located over the first die **110** and third die **130**. In at least one embodiment, the second die **120** may be positioned so that the second die **120** is located over an entirety of the first die **110** and an entirety of the third die **130** (see FIG. **1**B). The second layer **20** may also be positioned so that the first seal ring **117-1** in the first die **110** is located within the second seal ring **117-2** in the second die **120** (see FIG. **1**B).

[0108] A bonding process may then be performed to bond the second layer **20** to the first layer **10** through the bonding layer **150**. The bonding process may form, for example, a hybrid bond in which bonding layer bonding pad **157** in the bonding layer **150** is bonded to the die bonding pad **107** in the first die **110** and the bonding layer **150** is bonded to the die bonding film **105** of the first die **110**. The bonding layer **150** may also be bonded to the first encapsulation layer **118** of the first layer **10**. The bonding process may be performed at room temperature (room-temperature bonding) or at elevated temperatures (thermal bonding) depending on the specific bonding technique used.

[0109] FIG. **2**F is an intermediate structure after removal of the second carrier substrate **202** according to one or more embodiments. The second carrier substrate **202** may be detached from the second bonding film **142**, for example, by deactivating the adhesive layer (not shown) adhering the second carrier substrate **202** to the second bonding film **142**. The adhesive layer may be deactivated, for example, by a thermal anneal at an elevated temperature or by exposing the adhesive layer to ultraviolet light. metal

[0110] FIG. **2**G is an intermediate structure including a multilayer opening **0120** in the passivation layer **138** and the second bonding film **142** according to one or more embodiments. The multilayer opening **0120** may be formed in the passivation layer **138** and the second bonding film **142** by a photolithographic process. The photolithographic process may be similar to the photolithographic process described above for forming the metal features **116** in the first die **110**. In the photolithographic process, a photoresist mask (not shown) may be formed on an upper surface of the second bonding film **142**. The photoresist mask may be photolithographically patterned to include an opening over the die bonding pad **107**. An etching process (e.g., wet etching, dry etching, etc.) may then be used to form the multilayer opening **0120** in the second bonding film **142** and the passivation layer **138** through the opening in the photoresist mask. The multilayer opening **0120** in the second bonding film **142** and the passivation layer **138** may be formed to expose an upper surface of the die bonding pad **107**. The photoresist mask may be subsequently removed by ashing, dissolving the photoresist mask or by consuming the photoresist mask during the etch process.

[0111] It should be noted that at this point (e.g., before a bump loop process), testing such as wafer acceptance testing (WAT) may be performed on the semiconductor device **100**. In that case, test openings similar to the multilayer openings **0120** may be formed in the intermediate structure of FIG. **2**G. The test openings may be formed in the second bonding film **142**, passivation layer **138** an die bonding film **105** to expose a surface of the contact pads **103** electrically coupled to (e.g., contacting) the test line portion metal features **116**T in the second die test line portion **180-2** of the second die **120**. One or more testing probes of a WAT tool may then be inserted into the test openings to contact the contact pads **103** and perform the testing of the semiconductor device. It should be noted that the timing of WAT testing is not necessarily limited but may be performed at other stages of the manufacturing process. Further details of the testing by the WAT tool are provided below (e.g., see FIG. **4**).

[0112] FIG. **2**H is an intermediate structure including the metal bump **190** on the die bonding pad **107** in the multilayer opening **0120** according to one or more embodiments. The metal bump **190** may include, for example, a C4 bump including a solder ball formed on the die bonding pad **107**. The metal bump **190** may be formed, for example, by one or more processes including ball mounting, electroplating, solder printing, solder immersion and solder injection. The metal bump **190** may contact the die bonding pad **107** and a sidewall of the multilayer opening **0120** in the passivation layer **138** and the second bonding film **142**. In at least one embodiment, one or more underbump metallization (UBM) layers (not shown) may be formed on the die bonding pad **107**. The metal bump **190** may then be formed so as to contact the die bonding pad **107** through the UBM layers.

[0113] After forming the metal bump **190**, a dicing process may be performed to separate the semiconductor device **100** from the other semiconductor devices formed on the first carrier substrate **102** (e.g., carrier wafer). The dicing process may result in the formation of the sidewall **100***a* of the semiconductor device **100** including the sidewall **128***a* of the second encapsulation layer **128**, the sidewall **118***a* of the first encapsulation layer **118***a* and the sidewall **102***a* of the first carrier substrate **102**.

[0114] FIG. **3** is a flow chart illustrating a method of forming a semiconductor device according to one or more embodiments. With reference to FIGS. **1**A, **21**, **5**A, **7**, and **8**, step **310** includes forming a first layer **10** including a first die **110** including a first die edge **110***a* and a first die seal ring **117-1**. Step **320** includes forming a second layer **20** including a second die **120** including a second die edge **120***a* and a second die seal ring **117-2**, wherein at least one of the first die **110** includes a first die test line portion **180-1** between the first die edge **110***a* and the first die seal ring **117-1**, or the second die **120** includes a second die test line portion **180-2** between the second die edge **120***a* and the second die seal ring **117-2**. Step **330** includes attaching the first layer **10** to the second layer **20** such that a backside **120***b* of the second die **120** is located opposite the first die **110**.

[0115] FIG. **4** is a schematic illustration of a method of testing the semiconductor device **100** according to one or more embodiments. In at least one embodiment, the testing of the semiconductor device **100** may utilize the second die test line portion **180-2** in the second die **120**. In at least one embodiment, the testing of the semiconductor device **100** may include wafer acceptance testing (WAT). In at least one embodiment, a wafer acceptance testing tool **400** may be used to perform the testing of the semiconductor device **100**.

[0116] Wafer acceptance testing may play an important role in semiconductor fabrication by helping to ensure the quality and reliability of semiconductor wafers before they proceed to subsequent manufacturing processes. It may be performed, for example, near the end of the semiconductor device fabrication process (e.g., wafer fabrication process) or after completion of the semiconductor device fabrication process.

[0117] An objective of wafer acceptance testing may include identifying any defects, faults, or deviations in the electrical and physical characteristics of the semiconductor device **100**. The testing may involve subjecting the semiconductor device **100** to a series of tests to assess its functionality, performance, and overall quality. The testing may be performed using the wafer acceptance testing tool **400** and test structures that are designed to measure various parameters and behaviors of one or more integrated circuits (ICs) in the first die **110** and second die **120** of the semiconductor device **100**.

[0118] The WAT tool **400** may perform, for example, electrical testing of the semiconductor device **100**. This may involve probing the individual ICs in the semiconductor device **100** to measure the ICs' electrical characteristics. Various tests may be performed by the WAT tool **400**, such as measuring current-voltage (IV) curves, checking for shorts and opens, verifying power supply voltages, and analyzing the functionality of different circuit elements.

[0119] The WAT tool **400** may also perform parametric testing to measure critical electrical parameters of the ICs, such as voltage levels, currents, timings, and capacitances. These tests may help ensure that the ICs meet the specified performance requirements.

[0120] The WAT tool **400** may also perform functional testing to evaluate the overall functionality of the ICs by subjecting them to specific input stimuli and verifying the expected output responses. This may help ensure that the ICs perform their intended functions correctly.

[0121] The WAT tool **400** may also perform reliability tests to assess the long-term durability and stability of the ICs under various operating conditions. These tests include temperature cycling, burn-in testing, electrostatic discharge (ESD) testing, and other stress tests to simulate real-world usage scenarios and identify potential failure mechanisms.

[0122] The WAT tool **400** may also perform defect detection to inspect the semiconductor device **100** for physical defects using techniques such as optical microscopy, scanning electron microscopy (SEM), and other non-destructive testing methods. Defects such as contamination, particles, scratches, or pattern irregularities may be identified and categorized by the WAT tool **400**.

[0123] The WAT tool **400** may also statistically analyze the data obtained by testing the semiconductor device **100** to determine the overall quality of a batch of the semiconductor devices **100**. Data from the batch of semiconductor devices **100** may be collected and analyzed to assess process variations, yield rates, and identify any systematic issues that need to be addressed. The batch of semiconductor devices **100** may be accepted or rejected based on the results of testing by the WAT tool **400**.

[0124] Referring again to FIG. **4**, in at least one embodiment, the WAT tool **400** may be implemented by a computer, server, etc. The WAT tool **400** may include a processing device **410** such as a central processing unit (CPU), microprocessor, etc. The WAT tool **400** may also include a memory device **420** (e.g., random access memory (RAM), read-only memory (ROM), etc.). The memory device **420** may store data and programs including instructions form performing various operations in the WAT tool **400**. The memory device **420** may also store data generated by the testing performed by the WAT tool **400**. The processing device **410** may access the data and programs in the memory device **420**, and execute the instructions in order to perform various methods including a method of testing (e.g., wafer acceptance testing) the semiconductor device. The processing device **410** may also store test data generated by the testing performed by the WAT tool **400** in the memory device **420**. The processing device **410** may also perform analysis on the test data of testing performed by the WAT tool **400**, generate test analysis data based on the analysis and store the test analysis data in the memory device **420**. The WAT tool **400** may also include a monitor **430** (e.g., display device) and the processing device **410** may generate a display signal for generating a display on the monitor **430** including the test data, test analysis data, etc.

[0125] The WAT tool **400** may also include an input/output (I/O) device **440** (e.g., signal transmitter/receiver) for transmitting a test signal generated by the processing device **410** or according to an instruction generated by the processing device **410**. The I/O device **440** may also receive a signal from the semiconductor device **100** in response to the test signal.

[0126] The WAT tool **400** may include one or more testing probes **402** connected to the I/O device **400** by a signal transmission cable **401**. Other forms of signal transmission, such as wireless forms may be used. A method of testing the semiconductor device **100** may include forming one or more test openings OT in the functional circuit portion **170** and second die test line portion **180-2** of the second die **120**. The test openings OT may be formed, for example, by performing a photolithographic process (similar to the photolithograpic process described above for forming the metal features **116** in the first die **110**). The test openings OT may include multilayer openings formed in the backside etch stop layer **113** and gap fill dielectric layer **104**. The test openings OT may be formed to expose a surface of the contact pads **103** electrically coupled to (e.g., contacting) the metal features **116** in the functional circuit portion **170** of the second die **120**. The test openings OT may also be formed to expose a surface of the contact pads **103** electrically coupled to (e.g., contacting) the test line portion metal features **116**T in the second die test line portion **180-2** of the second die **120**.

[0127] The probes **402** may then be inserted into the test openings OT so as to contact one or more of the exposed contact pads **103** in the second die test line portion **180-2**. The probes **402** may electrically couple the WAT tool **400** to the contact pads **103** through the cables **401**. The WAT tool **400** may then transmit a test signal through a cable **401** and probe **402** to the exposed contact pad **103**. The test signal may be transmitted from the contact pad **103** through an integrated circuit included in the first die **110** and second die **120**. Another probe **402** contacting another exposed contact pad **103** may detect a response to the test signal and transmit the response through another cable **401** to the WAT tool **400**.

[0128] FIG. **5**A is a vertical cross-sectional view of the semiconductor device **100** having a first alternative design according to one or more embodiments. As illustrated in FIG. **5**A, the first alternative design of the semiconductor device **100** may be substantially similar to the original design in FIG. **1**A. However, in the first alternative design, the semiconductor device **100** includes a first die test line portion **180-1** in the first die **110**. The first die test line portion **180-1** may be adjacent the functional circuit portion **170** in the first die **110**. The first die test line portion **180-1** may be substantially similar to the second die test line portion **180-2** in the original design in FIG. **1**A.

[0129] To accommodate the first die test line portion **180-1**, the length of the first die **110** in the x-direction may be greater than the length of the first die **110** in the x-direction in the original design in FIG. **1**A. The length of the third die **130** in the x-direction may also be greater than the length of the third die **130** in the x-direction in the original design in FIG. **1**A.

[0130] The first die test line portion **180-1** may be accessed during wafer acceptance testing through the second die **120**. To allow access to the first die test line portion **180-1**, the first die test line portion **180-1** may include a test line portion bonding pad **107**T in the die bonding film **105** of the first die **110**. The test line portion bonding pad **107**T may be substantially similar to the bonding pad **107** in the die bonding film **105** of the first die **110**. The semiconductor device **100** may also include a bonding layer bonding pad **657** in the bonding layer **150**. The bonding layer bonding pad **657** may contact the test line portion bonding pad **107**T. The bonding layer bonding pad **657** may be substantially similar to the bonding layer bonding pad **157** in the bonding layer **150**. The second die **120** may also include a TSV **629** electrically coupled to the bonding layer bonding pad **657** in the bonding layer **150** and the metal features **116** in the second die **120**. The TSV **629** may be substantially similar to the TSV **129** in the second die **120**. The TSV **629** may be substantially similar to the TSV **129** in the second die **120**. The first die test line portion **180-1** may thereby be electrically coupled to the second die (e.g., a functional circuit area of the second die **120**).

[0131] The first die test line portion **180-1** in the first die **110** may, therefore, be used as a test line. In particular, the first die test line portion **180-1** may be used to monitor the influence of the process on the semiconductor device **100** having the first design and the integrated circuit in the semiconductor device.

[0132] In the first alternative design, the semiconductor device **100** may include a connecting structure including the test line portion bonding pad **607**, the bonding layer bonding pad **657** in the bonding layer **150** and the TSV **629** in the second die **120**. However, it should be noted that the semiconductor device **100** may alternatively include a plurality of the connecting structures between the first die test line portion **180-1** and the second die **120**. That is, the semiconductor device **100** may include a plurality of test line portion bonding pads **607**, bonding pads **657** and TSVs **629**.

[0133] FIG. **5**B is a schematic illustration of a method of testing the semiconductor device **100** having the first alternative design according to one or more embodiments. The method of testing the semiconductor device **100** having the first alternative design may be substantially similar to the method of testing the semiconductor device **100** having the original design in FIG. **1**A (see FIG. **4**). In at least one embodiment, the testing of the semiconductor device **100** may utilize the first die test line portion **180-1** in the first die **110**. In at least one embodiment, the wafer acceptance testing tool **400** may be used to perform the testing of the semiconductor device **100**.

[0134] As illustrated in FIG. **5**B, the probe **402** may be inserted into the test opening OT so as to contact an exposed contact pad **103** electrically coupled to the TSV **629**. The probe **402** may electrically couple the WAT tool **400** to the contact pad **103** through the cable **401**. The WAT tool **400** may then transmit a test signal through a cable **401** and probe **402** to the exposed contact pad **103**. The test signal may be transmitted from the contact pad **103** through an integrated circuit included in the first die **110** and second die **120**. The probe **402** may detect a response to the test signal and transmit the response through cable **401** to the WAT tool **400**.

[0135] FIG. **6** is a flow chart illustrating a method of testing the semiconductor device **100** according to one or more embodiments. Step **610** includes exposing a contact pad electrically coupled to a test line portion in a die in the semiconductor device. Step **620** includes contacting a probe of a WAT tool to the exposed contact pad. Step **630** includes transmitting a test signal from the WAT tool to the probe. Step **640** includes transmitting a response to the test signal from the probe to the WAT tool.

[0136] FIG. **7** is a vertical cross-sectional view of the semiconductor device **100** having a second alternative design according to one or more embodiments. As illustrated in FIG. **7**, the second alternative design of the semiconductor device **100** may be substantially similar to the first alternative design in FIG. **5**A. In particular, the first die **110** may include the first die test line portion **180-1**.

[0137] However, in the second alternative design, the semiconductor device **100** may not include the connecting structure between the first die test line portion **180-1** and the second die **120**. That is, the semiconductor device **100** may not include the test line portion bonding pad **607**, bonding layer bonding pad **657** and TSV **629** that may be included in the first alternative design in FIG. **5**A.

[0138] That is, the first die test line portion **180-1** may not be electrically coupled to the second die **120**. In this case, the first die test line portion **180-1** may not be used for testing, but may be used, for example, as a heat dissipation structure. The first die test line portion **180-1** may also be used to reduce the area of the third die **130**. In at least one embodiment, the first die test line portion **180-1** may completely eliminate the need for the third die **130**. This may help to reduce cost and balance stress (e.g., stress caused by mismatched coefficient of thermal expansions (CTEs)) in the level of the first die (e.g., same level, same structure).

[0139] FIG. **8** is a vertical cross-sectional view of the semiconductor device **100** having a third alternative design according to one or more embodiments. As illustrated in FIG. **8**, the third alternative design of the semiconductor device **100** may include both the first die test line portion **180-1** in the first die **110** and the second die test line portion **180-2** in the second die **120**. Further, the semiconductor device **100** may include the connecting structure between the first die test line portion **180-1** and the second die **120** as in the first alternative design in FIG. **5**A. That is, the semiconductor device **100** may include the test line portion bonding pad **607**, bonding layer bonding pad **657** and TSV **629**. Thus, the first die test line portion **180-1** may be electrically coupled to the second die **120** through the connecting structure.

[0140] The third alternative design in FIG. **8** may allow for separate monitoring of the first die **110** and the second die **120**. In particular, this design may allow the first die **110** and the second die **120** to be easily analyzed by separate testing by the WAT tool **400** (see FIGS. **4** and **5**B). Separate monitoring may be especially advantageous where the first die **110** and the second die **120** are formed in different processes. Separate monitoring may improve the electrical die sorting (EDS) process and may affect the affordability of the semiconductor device **100**.

[0141] Referring to FIGS. **1**A-**8**, a semiconductor device **100** may include a first die **110** including a first die edge **110***a* and a first die seal ring **117-1**, a second die **120** bonded to the first die **110** and including a second die edge **120***a* and a second die seal ring **117-2**, and a test line portion including at least one of a first die test line portion **180-1** in the first die **110** between the first die edge **110***a* and the first die seal ring **117-1**, or a second die test line portion **180-2** in the second die **120** between the second die edge **120***a* and the second die seal ring **117-2**. The test line portion may include the second die test line portion **180-2**, and the second die test line portion **180-2** may include a test line portion metal feature **116**T configured to be electrically coupled to a testing tool. The second die test line portion **180-2** may further include a test line portion seal ring **117**T around the test line portion metal feature **116**T. The second die test line portion **180-2** may be between the second die edge **120***a* and the second die seal ring **117-2** in a first direction, and a distance between the second die edge **120***a* and the second die seal ring **117-2** in the first direction may be greater than a distance between the second die edge **120***a* and the second die seal ring **117-2** in a second direction perpendicular to the first direction. The first die seal ring **117-1** may have a first die seal ring length in the first direction and a first die seal ring width less than the first die seal ring length in the second direction. The second die seal ring **117-2** may have a second die seal ring length in the first direction and a second die seal ring width less than the second die seal ring length in the second direction, and the second die seal ring length may be greater than the first die seal ring length. The second die **120** may further include a die substrate **108**, and a via **129** in the die substrate **108** and configured to electrically coupled the second die **120** to the first die **110**. The semiconductor device **100** may further include a bonding layer **150** between the first die **110** and the second die **120**, and a bonding layer bonding pad **157** in the bonding layer **150** and bonded to the via **129**. The first die **110** may include a die bonding film **105**, and a die bonding pad **107** in the die bonding film **105**, wherein the die bonding pad **107** may be bonded to the bonding layer bonding pad **157**. The semiconductor device **100** may further include a dummy die **130** adjacent the first die **110** and bonded to the second die **120** by the bonding layer **150**, and a first encapsulation layer **118** encapsulating the dummy die **130** and the first die **110**. The semiconductor device **100** may further include a second encapsulation layer **128** encapsulating the second die **120**, wherein the second encapsulation layer **128** may be bonded to the first encapsulation layer **118** by the bonding layer **150**. The test line portion may include the first die test line portion **180-1**, and the first die test line portion **180-1** may include a test line portion metal feature **116**T configured to be electrically coupled to a testing tool through the second die **120**. The second die **120** may further include a die substrate **108**, and a via **129** in the die substrate **108** and configured to electrically couple the second die **120** to the first die test line portion **180-1**. The first die test line portion **180-1** may be electrically isolated from the second die **120**. The test line portion may include the first die test line portion **180-1** and the second die test line portion **180-2**.

[0142] Referring again to FIGS. **1**A-**8**, a method of making a semiconductor device **100** may include forming a first layer **10** including a first die **110** including a first die edge **110***a* and a first die seal ring **117-1**, forming a second layer **20** including a second die **120** including a second die edge **120***a* and a second die seal ring **117-2**, wherein at least one of the first die **110** includes a first die test line portion **180-1** between the first die edge **110***a* and the first die seal ring **117-1**, or the second die **120** includes a second die test line portion **180-2** between the second die edge **120***a* and the second die seal ring **117-2**, attaching the first layer **10** to the second layer **20** such that a backside **120***b* of the second die **120** is located opposite the first die **110**. The attaching of the first layer **10** to the second layer **20** may include attaching the first layer **10** to the second layer **20** such that the first die **110** is electrically coupled to a through silicon via (TSV) **129** in the second die **120**. The second die **120** may include the second die test line portion **180-2**, the second die test line portion **180-2** may include a test line portion metal feature **116**T and the forming of the second layer **20** may include forming the second layer **20** such that the test line portion metal feature **116**T is accessible through an opening in a passivation layer **138** of the second die **120**. The first die **110** may include the first die test line portion **180-1**, and the attaching of the first layer **10** to the second layer **20** may include attaching the first layer **10** to the second layer **20** such that the first die test line portion **180-1** is electrically coupled to a through silicon via (TSV) **629** in the second die **120**.

[0143] Referring again to FIGS. **1**A-**8**, a testing method may include providing a semiconductor device **100** including a first die **110** including a first die edge **110***a* and a first die seal ring **117-1**, a second die **120** bonded to the first die **110** and including a second die edge **120***a* and a second die seal ring **117-2**, and a test line portion including at least one of a first die test line portion **180-1** in the first die **110** between the first die edge **110***a* and the first die seal ring **117-1**, or a second die test line portion **180-2** in the second die **120** between the second die edge **120***a* and the second die seal ring **117-2**, etching the second die **120** to expose a contact pad **103** electrically coupled to a metal feature **116** in the second die **120**, and attaching a test probe **402** of a testing tool **400** to the exposed contact pad **103** to electrically couple the testing tool **400** to the test line portion.

[0144] The various embodiments disclosed herein provide a semiconductor device **100** that may include easier access to enable testing of various functions or portions of the semiconductor dies **110** and/or **130**. By providing a test line portion (**180-1**/**180-2**) on either of the first die **110** or second die **120**, the semiconductor device **100** design may be more flexible and may increase the ability to monitor the semiconductor device **100** during production. The test line portion (**180-1**/**180-2**) may allow a WAT tool **400** to test various functions or portions of the semiconductor dies **110**, **120** as well as determine whether physical stresses or thermal expansion during the manufacturing process degrades or damages the semiconductor device **100**. This may allow the WAT tool **400** to monitor the influence of the manufacturing process on the semiconductor device **100**. Various embodiments are disclosed to form the semiconductor device **100** with stacked semiconductor dies **110**, **120**, wherein at least one of the stacked dies **110**, **120** includes a test line portion (**180-1**/**180-2**). In addition, various testing methods are disclosed that may utilize the test line portion (**180-1**/**180-2**) to test various functions and/or structures of the semiconductor device **100**.

[0145] The foregoing outlines features of several embodiments or examples so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments or examples introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor device comprising: a first die including a first die edge and a first die seal ring; a second die bonded to the first die and including a second die edge and a second die seal ring; and a test line portion comprising at least one of: a first die test line portion in the first die between the first die edge and the first die seal ring; or a second die test line portion in the second die between the second die edge and the second die seal ring.
2. The semiconductor device of claim 1, wherein the test line portion includes the second die test line portion, and the second die test line portion comprises a test line portion metal feature configured to be electrically coupled to a testing tool.
3. The semiconductor device of claim 2, wherein the second die test line portion further comprises a test line portion seal ring around the test line portion metal feature.
4. The semiconductor device of claim 2, wherein the second die test line portion is between the second die edge and the second die seal ring in a first direction, and a distance between the second die edge and the second die seal ring in the first direction is greater than a distance between the second die edge and the second die seal ring in a second direction perpendicular to the first direction.
5. The semiconductor device of claim 4, wherein the first die seal ring has a first die seal ring length in the first direction and a first die seal ring width less than the first die seal ring length in the second direction.
6. The semiconductor device of claim 5, wherein the second die seal ring has a second die seal ring length in the first direction and a second die seal ring width less than the second die seal ring length in the second direction, and the second die seal ring length is greater than the first die seal ring length.
7. The semiconductor device of claim 2, wherein the second die further comprises: a die substrate; and a via in the die substrate and configured to electrically couple the second die to the first die.
8. The semiconductor device of claim 7, further comprising: a bonding layer between the first die and the second die; and a bonding layer bonding pad in the bonding layer and bonded to the via.
9. The semiconductor device of claim 8, wherein the first die comprises: a die bonding film; and a die bonding pad in the die bonding film, wherein the die bonding pad is bonded to the bonding layer bonding pad.
10. The semiconductor device of claim 8, further comprising: a dummy die adjacent the first die and bonded to the second die by the bonding layer; and a first encapsulation layer encapsulating the dummy die and the first die.
11. The semiconductor device of claim 10, further comprising: a second encapsulation layer encapsulating the second die, wherein the second encapsulation layer is bonded to the first encapsulation layer by the bonding layer.
12. The semiconductor device of claim 1, wherein the test line portion includes the first die test line portion, and the first die test line portion comprises a test line portion metal feature configured to be electrically coupled to a testing tool through the second die.
13. The semiconductor device of claim 12, wherein the second die further comprises: a die substrate; and a via in the die substrate and configured to electrically couple the second die to the first die test line portion.
14. The semiconductor device of claim 12, wherein the first die test line portion is electrically isolated from the second die.
15. The semiconductor device of claim 1, wherein the test line portion includes the first die test line portion and the second die test line portion.
16. A method of making a semiconductor device, the method comprising: forming a first layer comprising a first die including a first die edge and a first die seal ring; forming a second layer comprising a second die including a second die edge and a second die seal ring, wherein at least one of: the first die comprises a first die test line portion between the first die edge and the first die

seal ring; or the second die comprises a second die test line portion between the second die edge and the second die seal ring; and attaching the first layer to the second layer such that a backside of the second die is located opposite the first die.

17. The method of claim 16, wherein the attaching of the first layer to the second layer comprises attaching the first layer to the second layer such that the first die is electrically coupled to a through silicon via (TSV) in the second die.

18. The method of claim 16, wherein the second die includes the second die test line portion, the second die test line portion comprises a test line portion metal feature and the forming of the second layer comprises forming the second layer such that the test line portion metal feature is accessible through an opening in a passivation layer of the second die.

19. The method of claim 16, wherein the first die includes the first die test line portion, and the attaching of the first layer to the second layer comprises attaching the first layer to the second layer such that the first die test line portion is electrically coupled to a through silicon via (TSV) in the second die.

20. A testing method, comprising: providing a semiconductor device comprising: a first die including a first die edge and a first die seal ring; a second die bonded to the first die and including a second die edge and a second die seal ring; and a test line portion comprising at least one of: a first die test line portion in the first die between the first die edge and the first die seal ring; or a second die test line portion in the second die between the second die edge and the second die seal ring; etching the second die to expose a contact pad electrically coupled to a metal feature in the second die; and attaching a test probe of a testing tool to the exposed contact pad to electrically couple the testing tool to the test line portion.
